- Fat helps the body to absorb vitamins A, D and E. These vitamins are fat-soluble, meaning they can only be absorbed with the help of fats.
- Any fat not used by body's cells or to create energy is converted into body fat. Likewise, unused carbohydrate and protein are also converted into body fat.
- All types of fat are high in energy. A gram of fat, whether saturated or unsaturated, provides 9kcal (37kJ) of energy compared with 4kcal (17kJ) for carbohydrate and protein. The fatty acid composition of oil is given in **table 5.1**.

5.3.2 Refined oils

Vegetable oils are produced from oil-containing seed, fruits or nuts by various pressing processes, by solvent extraction and also by combination of these. A seed cake that is relatively high in protein remains, after fat extraction is often used for animal feed.

5.3.3. Hydrogenation – vanaspathi and margarine

Hydrogenation

Plant oils contain a large percent of unsaturated fatty acids and hence have a tendency to become rancid. These unsaturated glycerides in oil can be converted to saturated glycerides by the addition of hydrogen. This process is known as **"hydrogenation"**.

Hydrogenated fat is manufactured from vegetable oils by the addition of molecular hydrogen to the double bonds in the unsaturated fatty acids in the presence of nickel.

Unsaturated Fatty Acids are of two types. There are

- MUFA – Mono Unsaturated Fatty Acids

PUFA – Poly Unsaturated Fatty Acids

DO YOU KNOW...?

As per the Heart Association of America, an individual should consume PUFA up to 10% and MUFA up to 15% of his/her total calories of the daily intake. MUFAs lower the level of bad cholesterol (LDL) in the blood and raise the good cholesterol (HDL). PUFAs reduce both the good and the bad cholesterol.

Vanaspathi

Hydrogenated oil in India is known as **"Vanaspathi"**. It is manufactured by hydrogenating refined groundnut oil or a mixture of groundnut oil with other edible vegetable oils. Good and Bad Fatty Acids are found in Vanaspathi.

DO YOU KNOW...?

According to Vanaspathi Control Order, the melting point of Vanaspathi should be between 31°C and 37° C and it should contain 5 percent sesame oil and should be fortified with vitamin A.

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